AI-Driven Crop Yield Prediction System

January Progress Report

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2 Introduction

2.1 Background and Motivation

Agricultural productivity faces increasing challenges from climate variability and environmental uncertainty. The growing global population and rising food demand require efficient farming practices to manage limited natural resources sustainably. Climate change has intensified extreme weather events such as droughts, floods, and unseasonal temperatures, creating greater uncertainty in crop production cycles.

In the UK, agriculture contributes approximately £10 billion annually to the economy, with wheat, barley, oats and oilseed rape representing major crops. Weather variability has a huge impact on yields. For example, the wet summer of 2012 caused wheat yields to drop nearly 19% below average, demonstrating how vulnerable UK agriculture is to climatic extremes.

Accurately predicting crop yield will enable farmers to optimise resource allocation, plan harvests, mitigate financial risks, and as a result improve their efficiency. Traditional methods rely on simple statistical approaches or farmer intuition, which most of the time fail to capture the complex interactions between environmental factors and crops. Machine learning offers the potential to model these non-linear relationships by learning from historical patterns in weather and yield data.

2.2 Project Aim

This project aims to develop and evaluate machine learning models that predict UK crop yields using historical weather data, identifying which environmental factors most strongly influence yield outcomes for different crop types and as a result helping farmers to make right decisions in order to improve their efficiency.

2.3 Project Objectives

1. Compile and integrate multi-source agricultural datasets covering crop yields and environmental factors
2. Perform exploratory analysis to identify relationships between weather conditions and yield outcomes
3. Feature Engineering to capture seasonal weather patterns aligned with crop growth stages

4. Develop and compare machine learning models including baseline (linear) and ensemble methods
5. Evaluate model performance and assess the relative importance of different predictors
6. Compare predictions across crop types (wheat, barley, oats, oilseed rape) and identify crop-specific patterns

2.4 Project Scope

This project focuses on four major UK crops: wheat, barley (winter and spring), oats, and oilseed rape (OSR). The analysis covers the period 2004-2024 using regional yield data from GOV.UK and weather data from the UK Met Office.

3 Technical Chapter: Methodology

3.1 Data Sources

Two primary data sources were integrated:

- **Yield Data:** Regional crop statistics (2004-2024) for wheat, barley, oats, and oilseed rape across four UK regions (England, Wales, Scotland, Northern Ireland)
- **Weather Data:** UK Met Office monthly records including temperature (max/min), rainfall, sunshine hours, frost days and humidity

The regional approach provides 84 observations per crop (4 regions $\times$ 21 years), compared to only 21 observations at national level.

3.2 Feature Engineering

Raw monthly weather data was transformed into 47 seasonal features aligned with crop growth stages:

- **Seasonal aggregations:** Winter (Dec-Feb), Spring (Mar-May), Summer (Jun-Aug), Autumn (Sep-Nov)
- **Critical period features:** Grain filling temperature, flowering sunshine, planting rainfall
- **Extreme indicators:** Heat stress days, frost days, drought or excess rain flags

This approach captures the biological reality that crops respond to weather during specific phenological windows, not uniformly throughout the year.

3.3 Modelling Approach

Train-Test Split: Temporal split with 2004-2019 for training (76%) and 2020-2024 for testing (24%).

Models Compared:

- Ridge Regression (linear baseline)
- Random Forest (non-linear, selected as primary model)

Random Forest was chosen for its ability to capture non-linear relationships, handle feature interactions, and provide interpretable feature importance scores. Another factor was the success it had on predicting crop yields on previous papers/researches.

Evaluation Metrics: R^2 (variance explained) and RMSE (error in t/ha).

3.4 Crop-Specific Modelling

Each crop required tailored feature selection based on its growth cycle:

Table 1: Selected features by crop

Crop	**Key Features**
Wheat	Winter sunshine, summer sunshine, grain filling rain
Winter Barley	Spring temperature, winter sunshine, planting rain, frost
Oats	Spring rain, flowering sunshine
OSR	Summer temperature, planting rain, winter frost

A key discovery was that spring and winter barley must be modelled separately due to fundamentally different growth cycles (1.44 t/ha yield difference).

3.5 Extreme Year Validation

Models were validated on two historically documented extreme weather years:

- **2010:** Coldest UK winter since 1978-79
- **2012:** Wettest English summer on record

Table 2: Extreme year validation results

Crop	**Year**	**Actual (t/ha)**	**Predicted (t/ha)**	**Error**
Wheat	2010	7.77	7.94	0.16
Wheat	2012	6.30	7.79	1.49
Barley	2010	5.62	5.71	0.09
Barley	2012	5.20	5.60	0.40
Oats	2010	5.60	5.31	0.29
Oats	2012	4.95	5.69	0.74
OSR	2010	3.48	3.49	0.01
OSR	2012	3.27	3.37	0.09

Models performed well on the 2010 cold winter (all errors <0.30 t/ha). The 2012 wet summer proved to be little bit more challenging, with wheat showing the largest error. In general, the model performed great on validating the results that these 2 extreme years have caused.

4 Progress and Current Results

4.1 Progress Made

- Expanded from national to regional modelling (21 → 84 observations per crop)
- Engineered 47 weather features aligned with crop phenology
- Developed Random Forest models for all five crop categories
- Discovered need to separate spring/winter barley
- Completed extreme year validation (2010, 2012)

4.2 Problems and Solutions

Problem	**Solution**
National models failed (insufficient data)	Regional disaggregation
Linear models poor performance	Random Forest captures non-linearity
Aggregated barley model failed (R^2=-0.16)	Separated spring/winter barley
OSR low R^2 despite optimization	Accepted as biological reality (pest-dominated)

4.3 Current Results

Table 3: Model performance (test set 2020-2024)

Crop	**R^2**	**RMSE (t/ha)**
Wheat	0.50	0.60
Winter Barley	0.45	0.48
Spring Barley	0.22	0.71
Oats	0.20	0.61
OSR	0.15	0.52

4.4 Key Findings

1. **Winter sunshine predicts wheat yield** (r=0.47 in regional analysis):an overlooked factor in farmer guidance

2. **Spring/winter barley are distinct**: 27% yield difference requires separate models
3. **OSR is pest-dominated, not weather-dominated**: only 15% variance explained by weather
4. **High winter frost correlates with better yields**: contrary to farmer intuition, likely due to vernalisation benefits
5. **Regional modelling is essential**: national aggregates mask important patterns

5 Revised Work Plan

5.1 January 2025 - Model Completion

- Finalise all crop-specific models
- Complete extreme year validation
- Document results and key findings
- Submit January progress report

Deliverable: January Deliverable

5.2 February 2025 - Interface Development

- Develop web-based prediction interface
- Integrate trained models for yield predictions
- Implement risk management calculator to help farmers

Deliverable: Working prediction interface prototype

5.3 March 2025 - Writing and Analysis

- Complete Results and Discussion chapters
- Improve the Sections / Add Depth
- Assemble first complete dissertation draft

Deliverable: First complete draft

5.4 April 2025 - Final Revisions

- Write Conclusions
- Final formatting and proofreading
- Prepare appendices

Final Submission: April, 2025

5.5 Summary of Deliverables

Deliverable	Deadline
January Progress Report	January, 2025
Prediction Interface Prototype	February, 2025
First Complete Draft	March, 2025
Final Dissertation	April, 2025